Exercise

Memory Forensics Exercise

This exercise introduces memory analysis. You can use the sources you have been provided with on Moodle to familiariase with Volatility.

Scenario

The system was captured while powered on following suspicious browsing activity.

Your task is to use **Volatility** to determine:

1. What operating system the system is running
2. What web browser process is active
3. What URL the browser was pointed at

You must document:

- Commands used
- Output snippets
- Reasoning

Part 1 – Identify the Operating System

Task

A. Determine:

- Operating system family
- Version
- Architecture (32-bit or 64-bit)

B. *Study the command before running it:*

```
vol -f memory.img windows.info
```

C. Answer the following questions:

- What OS is identified? ……

- Is the system 32-bit or 64-bit? ……

- What is the system build number? ……

Part 2 – Identify the Running Web Browser

Task

A. Identify any active browser processes.

B. Study the commands before running them:

```
-  vol -f memory.img windows.pslist
-  vol -f memory.img windows.psscan
```

C. Answer the following questions:

- Which browser process is running? ……

- What is the PID? ……

- What is the parent process? ……

- Does the parent-child relationship appear normal? ……

Part 3 – Identify URL

Task

A. Extract evidence of the website being accessed.

B. *There are different options to identify the website. Study this command you could use:*

Option - Dump Browser Process Memory

```
vol -f memory.img windows.cmdline
```

- What URL(s) are visible? ……

- Are they complete URLs or fragments? ……

- How confident are you that this was actively loaded? ……

- Could this be cached or background content? ……

- Identify an alternative command that would give you the same result

  ……

Part 4 – Corroboration

Task

Try to answer the following questions:

- Is there an active network connection? ……

- Does the remote IP correspond to the identified URL? ……